

Causal Inference and ML: Simpson's Paradox in Drug Recovery

Analyze the effect of a drug on recovery and demonstrate Simpson's paradox caused by confounding variables

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Github Repository:

<https://github.com/polinato/simpsons-paradox-drug-recovery>

Research Questions

Prediction

Can we accurately predict patient recovery outcome from clinical and demographic features?

Risk Optimization

Which patients benefit most or least from the drug, and how can we identify high-risk or high-response subgroups?

Causal Inference

Does the drug causally improve recovery after accounting for confounding factors?

Data Structure

Dataset

800 patients

Treatment: drug (1/0)

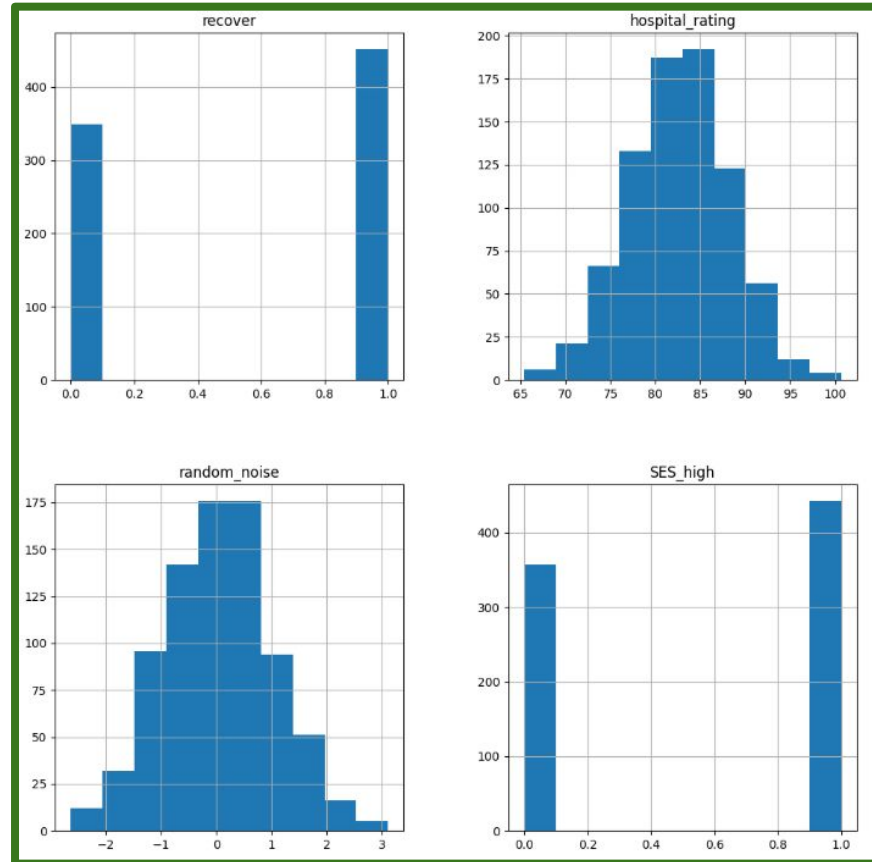
Outcome: recover (1/0)

Confounders: age,
SES, hospital_rating

	drug	age	recover	SES	hospital_rating	random_noise
0	Yes	Old	1	High	90.428932	1.554160
1	Yes	Old	1	High	85.714715	1.563413
2	Yes	Young	0	High	75.507660	1.642673
3	Yes	Old	1	Low	78.399887	-1.111967
4	Yes	Young	0	High	85.429478	-0.328618



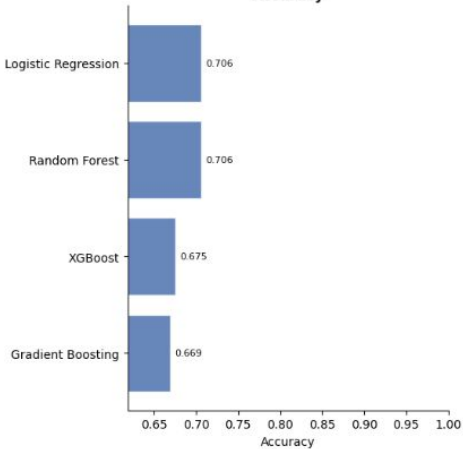
Data Structure - Visualization



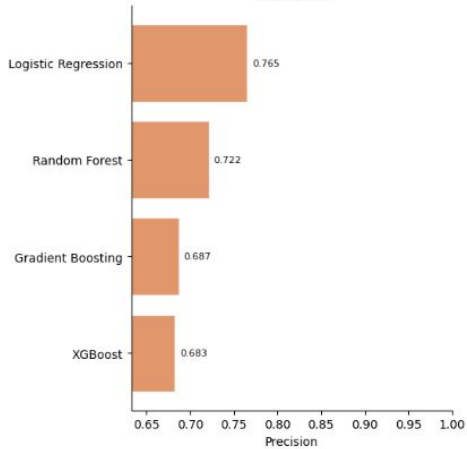
Regression Models - Results

Classification Model Comparison - Recovery Prediction

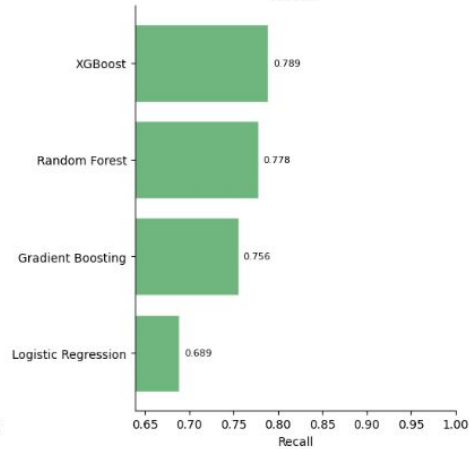
Accuracy



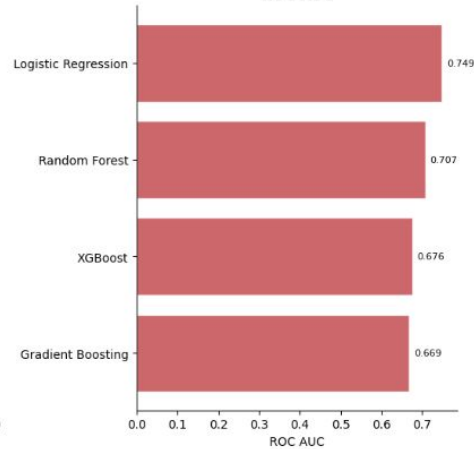
Precision



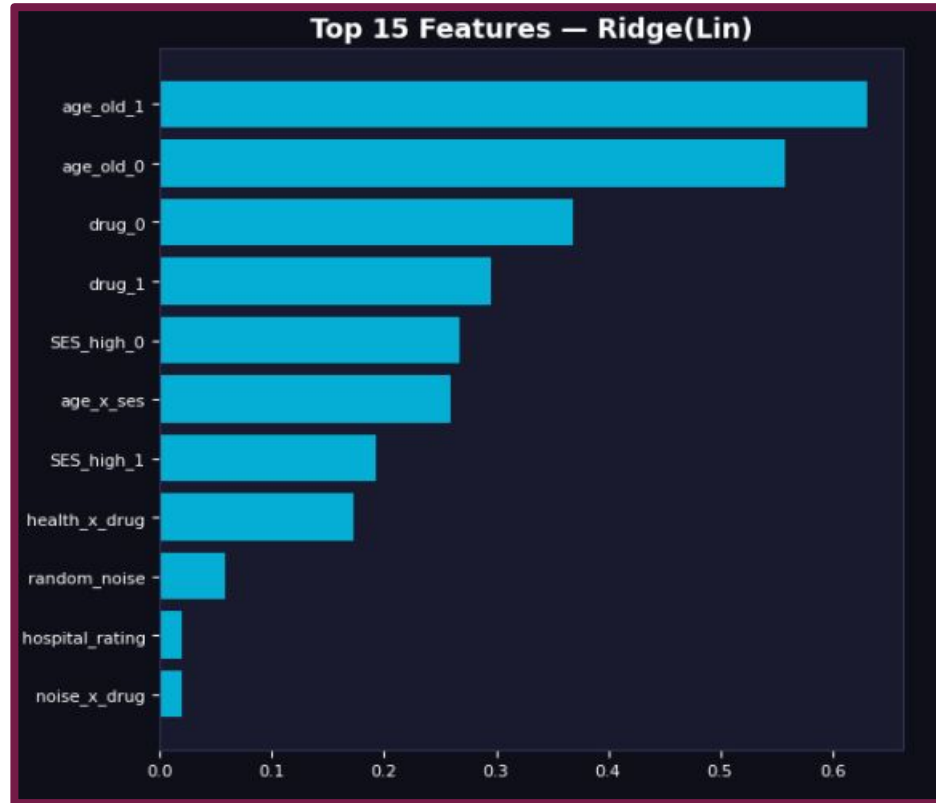
Recall



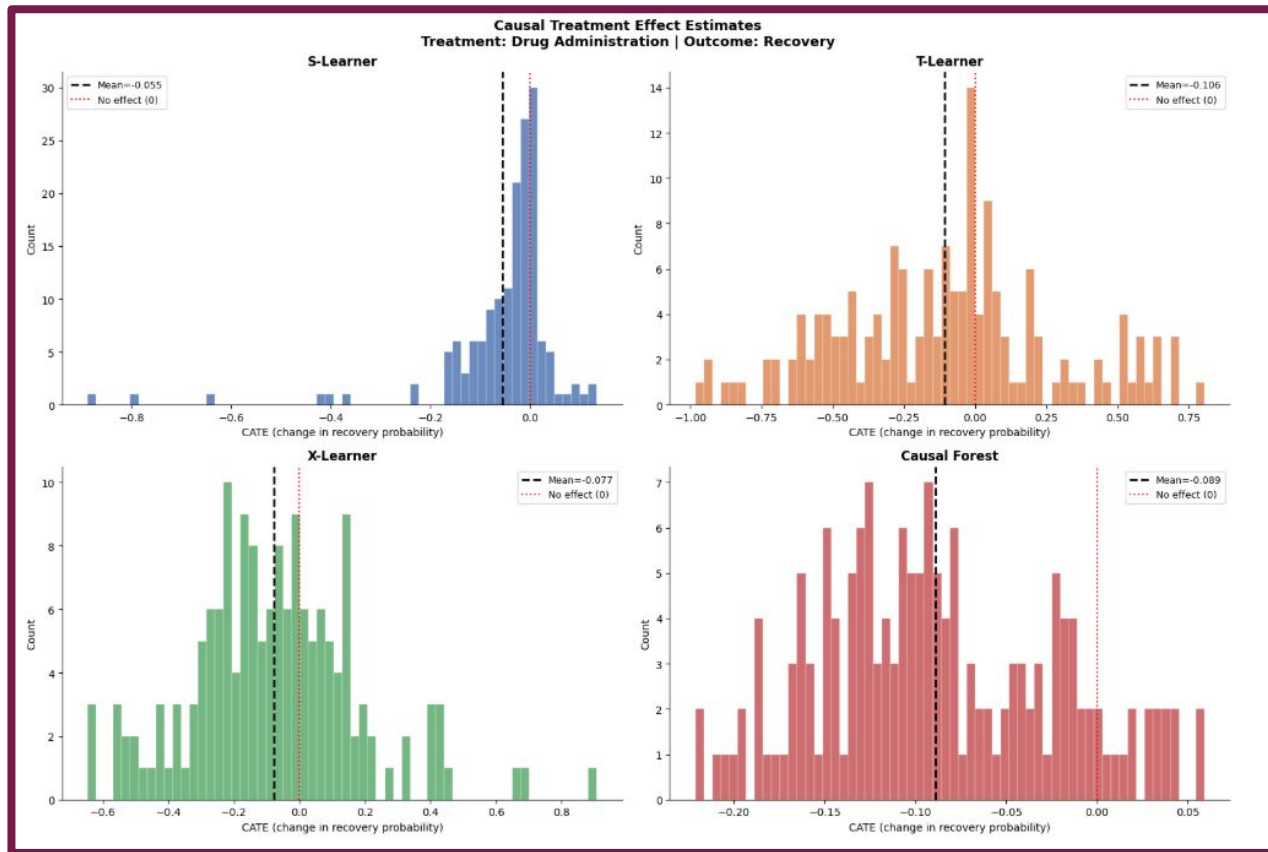
ROC AUC



Regression Models - Feature Importance

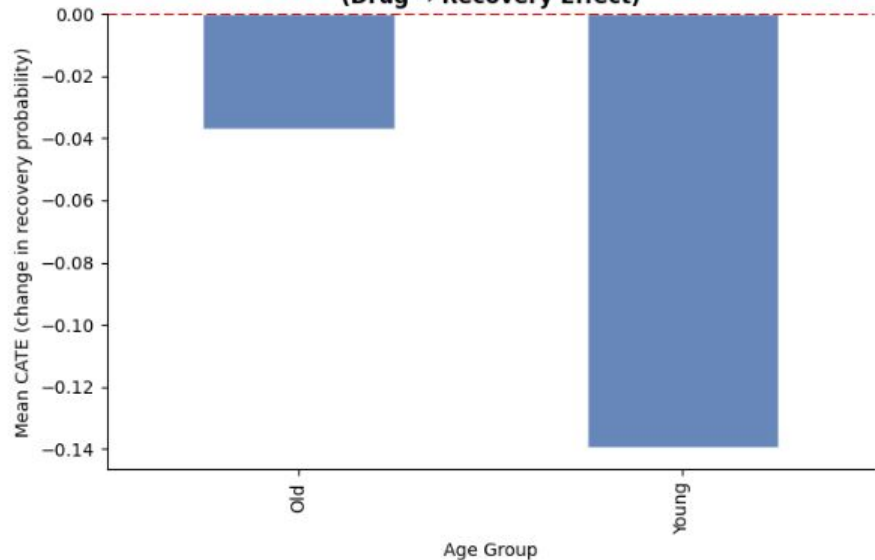


T,S,X-Learners and Causal Forests

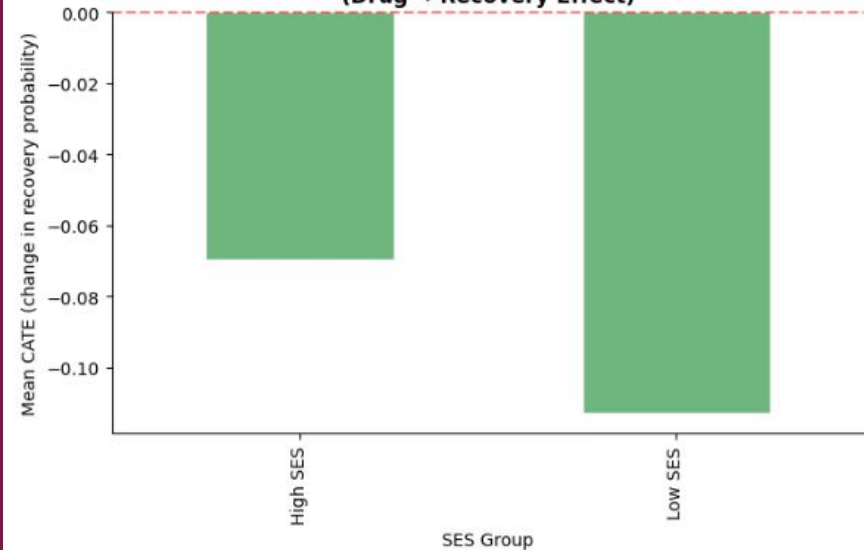


T,S,X-Learners and Causal Forests

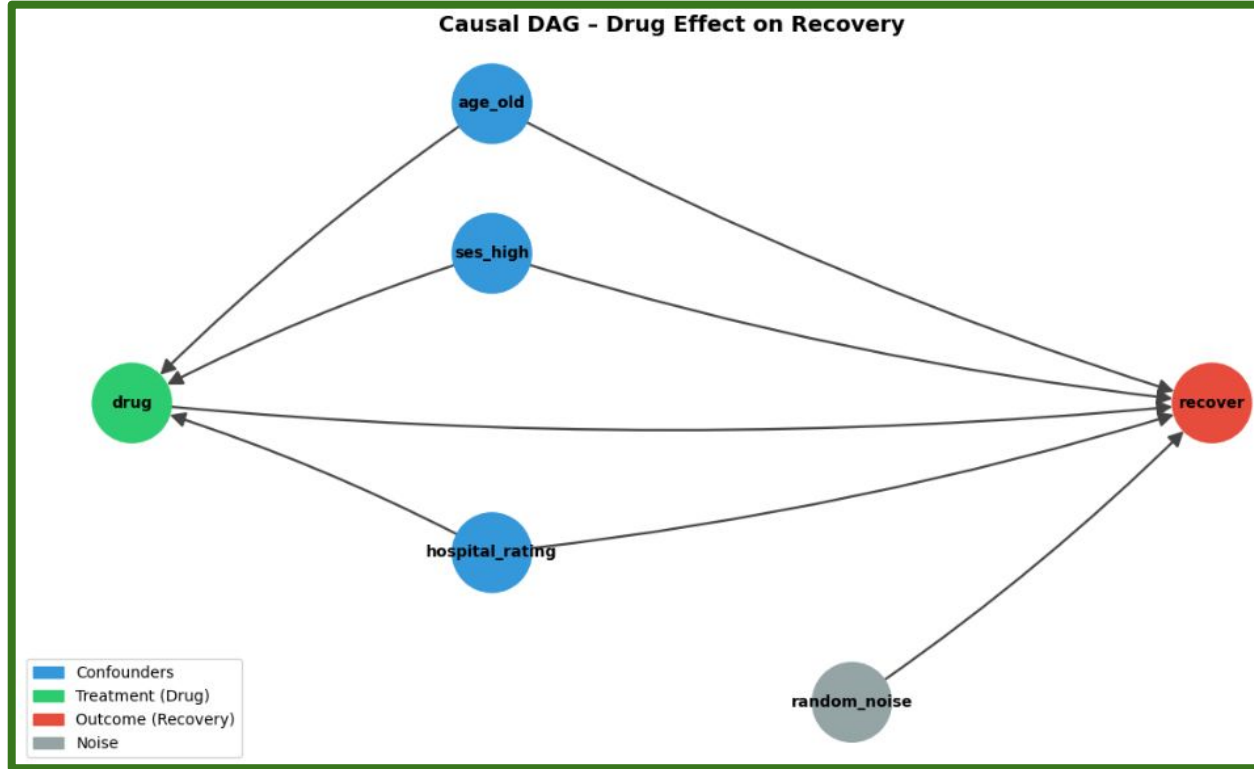
**Causal Forest CATE by Age Group
(Drug → Recovery Effect)**



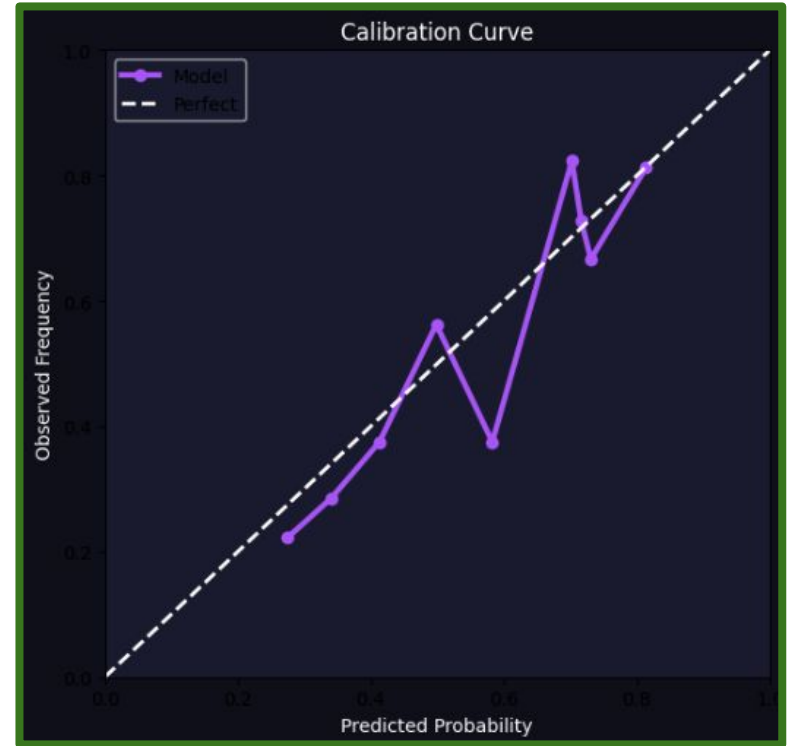
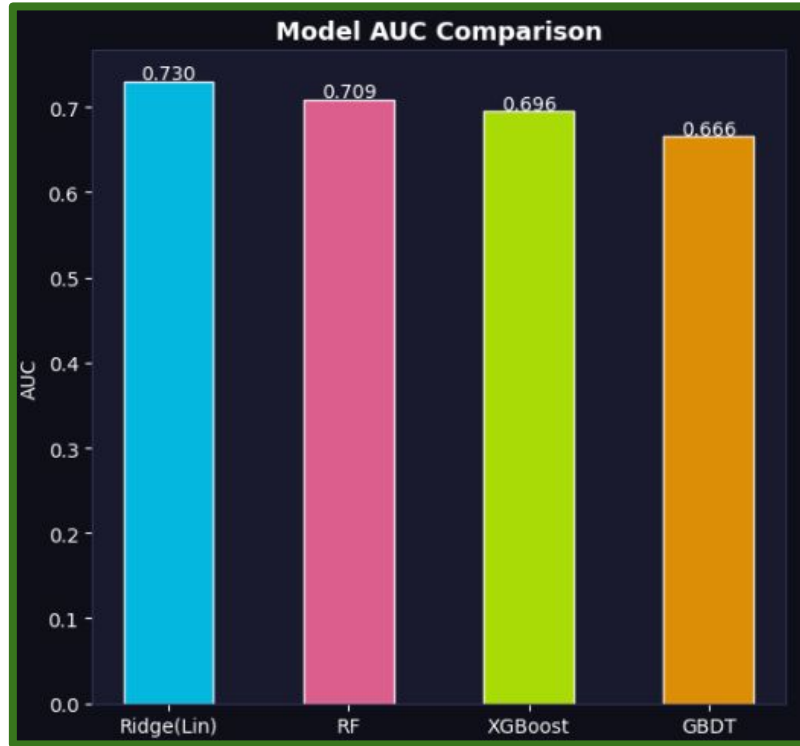
**Causal Forest CATE by SES Group
(Drug → Recovery Effect)**



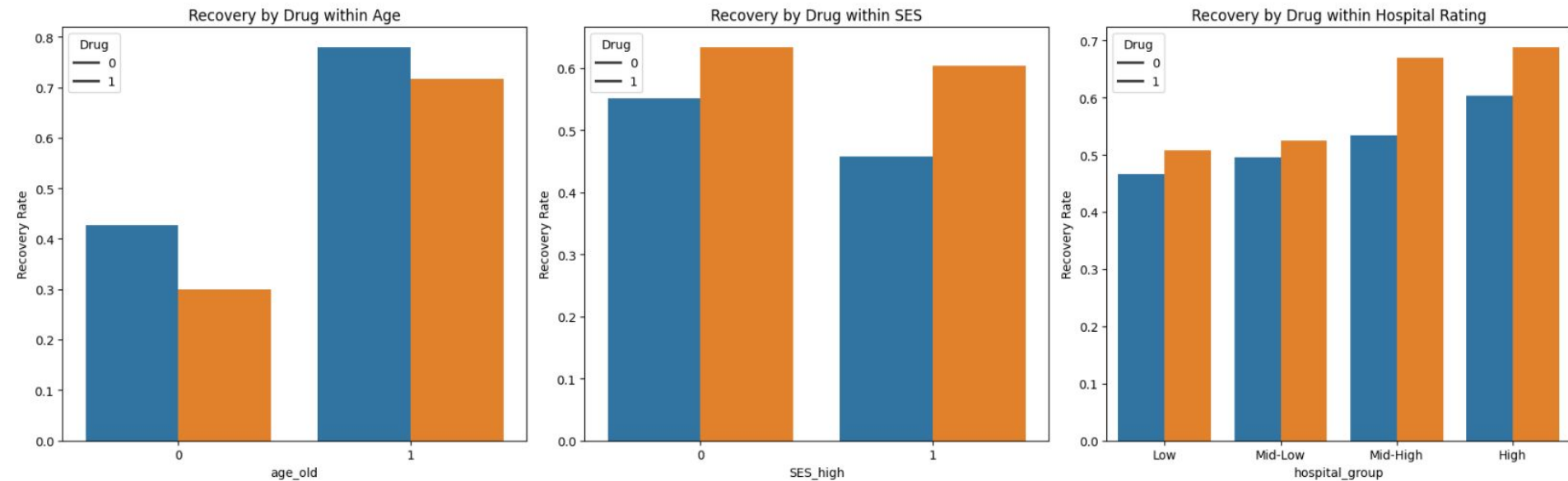
DAG - Directed Acyclic Graph



RQ - Prediction



RQ - Risk Optimization



RQ - Causal Inference

